Robot Design Speech Worksheet, 2026-11-05

Team: Look it's a Giraffe! | Tournament: Saturday, November 28

Part 1 is the fill-in-the-blank speech template. **Part 2** is the reference material to pull answers from: past meeting notes, the robot decision log, and the worksheet results.

Part 1: Speech template

How judging works (read this first)

Step	Time	What happens
Robot Design: Team Explanation	**5 minutes**	**This speech.** We talk, judges listen. No game field.
Robot Design: Question and Answer	5 minutes	Judges ask about what we said.

Five minutes is about **600 to 700 spoken words**. Speak slowly. Judges score only what they **hear**.

Judges tick one box per row. To be marked **Accomplished (3)** we need *clear evidence*. To be marked **Exceeds (4)** the judges must write down how we exceeded. Give them something to write.

Rubric area	What judges listen for at **Accomplished**	How to give them more (**Exceeds**)
IDENTIFY	Clear mission strategy. Clear use of building and coding resources.	Say *why* we chose missions in that order. Name the specific resources and what we took from each. Say who we asked for help.
DESIGN	*All* team members contributed ideas. Building and coding skills in *all* members.	Name **every** teammate and one thing each did. Show that people learned a new skill (building *and* coding).
CREATE	Clear explanation of innovative attachments and their purpose. Clear explanation of innovative code and sensor use.	Say what we **changed from the original** and why. Explain the code step by step. Say what each sensor does for us.
ITERATE	Clear evidence of **repeated** testing. Clear evidence of improvements **based on testing**.	Use **numbers** (how many runs, how far off). Tell the story: *we tested, we found a problem, we changed this, we tested again, it got better.*
COMMUNICATE	Detailed explanation of the process and lessons learned. Clear pride and enthusiasm.	Say what we would tell another team. Smile, pace yourselves, and celebrate what we are proud of.

Core Values count too. Teamwork, discovery, inclusion, innovation, impact and fun are all judged. Slip them in naturally (there is a prompt in section 7).

Rules for writing your part

1. Every blank needs a **fact from Part 2**. If you can't find it, write **[ASK: who]** and we will track it down.
2. Use **numbers** wherever you can ("10 runs", "within 2 mm"), not "pretty good" or "a lot".
3. Always answer **why**. A choice with no reason scores low.
4. Write it the way you talk. Short sentences. Read it out loud once you are done.
5. Don't read a list. Tell the story.

Who says what

Everyone speaks. Write the name of the person who will say each section. The times add up to about 4 minutes 40 seconds, which leaves a little room for pauses.

Section	Time	Speaker
1. Opening and mission strategy	0:30	
2. Choosing and building our robot	0:40	
3. Our attachments	1:00	
4. Our code and sensors	0:50	
5. Testing and improving	0:50	
6. What we learned	0:25	
7. Teamwork and closing	0:25	

You can split a section between two people. Make sure **all 9 of us** speak: Sylvie, Brynn, Jack, Fedor, Ira, Jordan, Ella, Alexis, Amelia.

Section 1. Opening and mission strategy (IDENTIFY)

Judges want: how we decided which missions to try, and who and what we learned from.

Hello, we are team _____ and today we will explain how we designed our robot.

Our robot game strategy is to attempt these missions: _____.

We chose those missions because _____.

We decided to do them in this order because _____.

(Think: which missions are worth the most points? Which are close together? Which can one attachment do? Mission 11 is worth 30 points. What is our plan for the others?)

To learn how to build and code, we used these resources: _____.

From _____ we learned _____, and we used it for _____.

When we got stuck, we asked _____ for help with _____.

(Think: bot-fluencers, the Bayview Glen conference sessions, the SPIKE app, build instructions. Name at least three and what each gave us.)

Section 2. Choosing and building our robot (IDENTIFY + DESIGN)

Judges want: a reason for our base robot, not just a name.

We looked at ____ different robot designs: _____.

We scored them on ____ things: _____.

We left out _____ on purpose, because _____.

The design that scored highest was _____ with _____ out of _____ points.

We chose it because _____ and _____.

(Think: the six scoring criteria and the totals are in the decision log, entries 1 and 2. What was the biggest reason, in your own words?)

After we built it, we noticed a problem: _____.

So we changed _____ to _____ because _____.

(Think: the front wheels. What was the problem with each version? Which one did we end up with, and why? Entries 3, 5 and 9.)

Section 3. Our attachments (CREATE)

Judges want: for each attachment, what it does, what we changed from the original, and why. Repeat the box below for each attachment you present.

We built ____ attachments. All of them attach to the robot's _____ using _____.

> **Attachment name:** _____
>
> It is used for mission(s): _____.
>
> It started as a design from _____.
>
> We changed _____ from _____ to _____. We did this because _____.
>
> One problem we found was _____. We fixed it by _____.
>
> It is good at _____. It is not so good at _____.

Repeat for: ☐ Lifting arm with seed bucket ☐ 2-axis arm with hook ☐ Forklift ☐ Mission 1 attachment ☐ Mission 11 basket

Our attachments swap on and off in about _____ seconds. We tried _____ pins to make it faster, but _____. We went back to _____ because _____.

We rated our attachments from 1 to 5 and ended up choosing _____ because _____.

*(Think: say **what is new**, not only what it is. A judge can read a build manual. They cannot know what we changed unless we say so.)*

Section 4. Our code and sensors (CREATE)

Judges want: a step-by-step explanation of how our code works and what our sensors do. Imagine explaining it to someone who has never coded.

We use ____ sensors on our robot: _____.

The sensor in the robot's _____ does _____. We use it to _____.

We added a second color sensor because _____.

Here is how squaring up on a line works, step by step:

1. First, _____.
2. Then, _____.
3. Finally, _____.

(Think: with one sensor it didn't work. With two sensors, what happens when one sees black?)

Our code uses a block that makes the robot drive straight, called _____.

It works like this: every 0.1 seconds the robot checks _____, and if it is off by a little, it _____.

The robot uses a number called **Kp** that decides _____, and a number called **Ki** that decides _____.

For turning we use _____ instead of motor degrees because _____.

(Think: the difference between a 90° motor turn and a 90° yaw turn. What does the gyro sensor measure? Entry 8 and the Kp/Ki section in Part 2.)

We also made the robot start slowly, speed up, and slow down at the end, because _____.

We changed the distance block from _____ to _____ because _____.

We figured out that _____ motor degrees equals _____ cm by _____.

Section 5. Testing and improving (ITERATE)

*This is the section where teams exceed. Judges want to hear **repeated** testing and **improvements because of the testing**. Use numbers.*

Tell **at least two** test stories using this pattern:

> **Story 1:** We wanted to know _____. So we tested by _____, _____ times. The results were _____ (give numbers). That told us _____. So we changed _____. After the change, we tested again and _____.
>
> **Story 2:** We wanted to know _____. So we tested by _____, _____ times. The results were _____ (give numbers). That told us _____. So we changed _____. After the change, we tested again and _____.

Ideas for stories (pick the ones we have the best evidence for):

- ☐ **Tuning Kp and Ki.** We tried Kp values of _____ and Ki values of _____, _____ runs each. We picked Kp _____ and Ki _____ because _____. Before, we were off by about _____ mm. After, we were off by about _____ mm.
- ☐ **Driving 1 meter, 10 times.** The results were _____.
- ☐ **The seed basket on Mission 11.** We tried _____ ideas. They caught the seed ____, ____ and ____ times out of 10. The best was _____.
- ☐ **Front wheel changes.** Original wheels, then mine cart wheels, then ball joint. The problem with each was _____.
- ☐ **Pins.** Axle pins versus quick pins.
- ☐ **Forklift tipping.** It tipped when _____. We fixed it by _____.
- ☐ **Squaring with two sensors.** Before: _____. After: _____.

We know some things are not finished. We still need to _____, and our plan is to _____.

(Tip: admitting what is not finished and having a plan for it sounds confident, not weak. Judges know robots are never done. See "gaps" in Part 2.)

Section 6. What we learned (COMMUNICATE)

Judges want: a detailed explanation of the process and lessons learned.

The biggest thing we learned about building a robot is _____.

The biggest thing we learned about coding is _____.

Something we thought would work, but didn't, was _____. We learned _____.

If we were giving advice to a new team, we would tell them _____.

If we had more time, we would _____.

(Think: mistakes are good here. A lesson that began as a mistake shows the judges we learned something.)

Section 7. Teamwork and closing (DESIGN + COMMUNICATE)

*Judges want: **all** team members contributed ideas, and skills in **all** team members. Plus pride and enthusiasm.*

Our team worked together in groups. Here is who did what:

Name	One thing I built or designed	One thing I coded, tested or learned
Sylvie		
Brynn		
Jack		
Fedor		
Ira		
Jordan		
Ella		
Alexis		
Amelia		

*(Everyone fills in their own row. If a row has only "built" and nothing about coding or testing, add something. The rubric asks for building **and** coding skills in all members. Who learned a new skill this season?)*

Core Values sentence (pick one or two and make them true for us):

- **Teamwork:** When _____ got stuck on _____, _____ helped by _____.
- **Discovery:** Something new we learned together was _____.
- **Inclusion:** We made sure everyone's ideas were heard by _____.
- **Innovation:** A creative way we solved a problem was _____.

- **Fun:** My favorite moment working on the robot was _____.

Closing:

The part of our robot we are proudest of is _____ because _____.

Thank you for listening. We would love to answer your questions about _____.

Checklist before we practice

- ☐ Every blank has a fact, a number or an **[ASK: who]**
- ☐ Every choice has a **because**
- ☐ All 9 teammates speak
- ☐ At least 2 test stories with numbers
- ☐ We explained what we **changed from the original** for each attachment
- ☐ We explained the code **step by step**
- ☐ We said what we learned, including a mistake
- ☐ A Core Value is in there
- ☐ We read it out loud, and it is under 5 minutes
- ☐ We can give the speech **without the field** and without reading from a page

Questions judges may ask (practice answers)

1. Why did you choose this base robot?
2. What is the most important change you made to your robot, and why?
3. How do you know your robot is accurate? How did you test it?
4. Walk us through how your code works.
5. What did you do when something didn't work?
6. Who built each attachment? Who wrote the code?
7. How did you decide which attachment to use for each mission?
8. What would you change if you had more time?
9. What is the hardest mission, and how does your robot do it?
10. What did each of you learn?

Part 2: Reference material

Use this to fill in the blanks. Facts come from our own notes, logs and worksheets. Items marked **[?]** or **(confirm)** are not yet certain.

A. What we are still missing

Judges are likely to ask about these. If you know the answer, tell Coach Chris so it can be written down.

Gap	Who probably knows
Mission 2 test run results, and the Move 1 and Move 2 distances in cm	Jordan, Fedor
Which Kp and Ki were in the code for the 1 m test, and which values are in the code now	Jordan
Ruler squaring check (back and each side)	Anyone, 5 minutes
Ten 90° turns accuracy test	Jordan, Jack
Seed order for the seed mission	Ira
How seeds come out of the seed bucket, and how it relates to the Mission 11 basket	Brynn, Jordan, Jack
Forklift and Mission 1 attachment pages (original build, what changed, results)	Jack, Ira, Ella, Amelia, Sylvie
Which Mission 1 attachment we chose, and why	Sylvie, Amelia
Where the second color sensor is mounted	Jack
Whether gears slip or have play on the 2-axis arm (the worksheet said "gears fall off")	Anyone who built it
The rest of the Kp/Ki attachment test sheets (seed bucket, Mission 1, the 10 confirm runs)	Whoever ran the tests
Who built which attachment ("we all collaborated" is not enough for judges)	Everyone
The ball caster drifting when accelerating or reversing ("it is being fixed")	Jordan

B. Meeting notes, organized by topic

B1. Choosing the robot

2026-09-15, bot-fluencers. Each of us presented an FLL YouTube creator and chose a first thing to try.

Presenter	Bot-fluencer	First to implement
Alexis	Team TurtleBots	Mission 01 from Max Points
Sylvie	Robotics Rules	Build a passive attachment
Brynn	Next Level Teacher	Starter Robot
Amelia	Giraffes #3388	Mission 1 attachment from the video
Jordan	Team TurtleBots + Next Level Teacher	(same as Alexis and Brynn)

Team name chosen: **Look it's a Giraffe!** (6 of 9 first-place votes).

2026-09-17, more bot-fluencers, and workstreams.

Presenter	Bot-fluencer	First to implement
Fedor, Ira	Platinum Pizzeria	Passive attachment for Mission 1
Jack	Mocking Bricks	Line following video
Ella	Robot Man	"Raise the platform" with a forklift

The team split into four groups: the starter bot (Sylvie, Brynn, Jack) and three Mission 1 attachments (Amelia and Ella from Giraffes #3388, Fedor and Ira from Platinum Pizzeria, Jordan and Alexis from Team TurtleBots).

The plan: build the starter bot, adapt each attachment to it, test all three and rate pros and cons, pick one for final refinement.

Starter bot research:

- **Robotics Rules StarterBot:** one SPIKE Prime set, gear-based attachment mount, $30.
- **Next Level Teacher HummerOne:** free with the FLL FastTrack course.
- Free alternatives: Pybricks 26-step StarterBot, FLL Tutorials base robot video.
- Team TurtleBots and Giraffes #3388 had no usable build instructions.

2026-09-22, scoring and decision. We scored four designs on **six criteria, all about how the robot performs in a match** (archetype match, attachment swap speed, squaring, stability under load, maneuverability, drive repeatability). Cost, build time and parts count were **deliberately left out** and treated as a separate decision.

Design	Score (of 30)	Rank
HummerOne	**23**	1
Comp Bot	20.5	2
GummyBears / Droid Bot M	20	3
StarterBot	14.5	4

HummerOne is the robot we are building. Drivetrain completed by Jack, Fedor, Jordan and Ira. The groups were reshuffled: Sylvie and Brynn moved onto attachments, and Fedor, Ira and Jordan moved onto the base robot. New bot-fluencers added: GummyBears Robotics (FLL Team #44355, 2022 Robot Design Award winners) and Zain Khan ("Coach Zain", creator of Comp Bot, with gyro turn and line following tutorials).

B2. Bayview Glen conference (2026-09-29)

Sessions we attended, and what we did about each. This is the clearest evidence of **using resources and seeking guidance**.

Session	Who	What we learned	What we did
How to Build a Functional Robot	Ira, Fedor	Square up against a wall. Shield color sensors from room light. **Smaller wheels wobble less and are more accurate. Accuracy matters more than speed.**	Recommended a ball caster for the front wheels (see B3). Judged our light shielding fine for now.
Programming 101	Brynn, Amelia	Types of blocks and turns (pivot, point/spin, arc). More confidence changing code.	
Programming 201	Jack, Jordan	Two-stage line following with one sensor. **Squaring up on a line needs two color sensors.**	We tested in the gym and **could not square up with one sensor**, so the team **added a second color sensor**.
PID	Jack, Jordan	Measure the error, multiply by a small number (**Kp**), and the robot corrects smoothly instead of zigzagging.	Led to Jordan's PID blocks (B3).
Programming 301	Jack	**Acceleration ramp** (start slow, speed up, slow down). **Yaw (gyro) turns**: turn to 90° of yaw, not 90° of motor, which only turns the wheel a quarter turn.	Both are in our code (B3).
Robot Game and Rules	Ira, Fedor, Brynn, Alexis, Amelia	Inspection bonus. Swap team members at the table. Referee can adjust some models. Seed mission order matters. Don't touch anything until the score is final.	
How to Prepare for a Judging Session	Ira, Fedor	**Pick a favorite mission. Explain how you solved it, what each person did, and how you used your time.** Always use the rubric. Describe things step by step. Be nice. "Don't trust AI": check real sources.	This worksheet.
Equity and Inclusion	Brynn, Amelia, Alexis	Inclusion versus equity. Help quieter teammates be heard.	
FIRST Like a Girl	Brynn, Amelia	Outreach for women and girls in STEM. ConnecTech FLL team.	

B3. Building and coding (2026-10-06 to 10-20)

2026-10-06. Jordan shared working **PID drive and turn My Blocks**. Jordan and Fedor wrote the **Mission 2** code with them and ran test runs (Kp 2.1, Ki 2 at the time). Jack and Ira worked on the forklift.

2026-10-13. Jordan's weekend work:

- Mission 2 and Mission 11 code progress.
- Front support: **mine cart wheels replaced with a Technic ball joint.** The wheels were very good in a straight line, but resisted when turning and stuck out past the sides. The ball joint turns better and stays

under the robot. It caused some **wobble**, so Jordan retuned Kp and Ki.

- **Gyro Move now takes cm** instead of motor degrees. Measured: **2070 motor degrees = 100 cm** with a 1 m measuring stick (**20.7 degrees per cm**).
- Mission 11 seed basket (Jordan, Brynn, Jack): catches the seed, but it sometimes goes over the top.
- Forklift (Ella, Amelia).

2026-10-20. Sylvie and Amelia worked on Mission 1. (The rest of the team did logo and innovation research.)

B4. Robot design work since then

- **2026-10-29 (robot day):** Tested the numbers that keep the robot steering straight (Kp and Ki), and found a setting that keeps it within a couple of millimetres on a 1 m drive. Started the same tests with each attachment. Everyone wrote down why we made each design choice. We rated our three attachments. The best seed basket idea caught the seed 7 out of 10 times.
- **2026-11-03:** Homework on the innovation project. Tournament details: **Saturday, November 28**, Western Technical Commercial School, pits open 7:30 AM.

B5. Judging tips we were given (use these!)

- Pick a favorite mission. Explain how we solved it, what each person did, and how we used our time.
- Always check the rubric.
- Walk through code **step by step**.
- Be encouraging to other teams and to our own team, and thank the judges on the way out.
- Plan to present **without the game field**.

C. Robot design decision log

Full log: robot_design/design-log.md. Summary, with the facts to use in a speech:

#	Date	Decision	Why, in one line
1	2026-09-17 to 09-22	HummerOne as the base robot	Free, a full-width attachment bar, and a perimeter frame that should square well
2	2026-09-22	Scored four designs; HummerOne 23/30	Highest on archetype match, attachment swap and stability, with no weak spot
3	Before 2026-10-01	Mine cart wheels replaced the original front wheels	Lower profile and less friction when turning, but not flush with the sides
4	2026-09-27 to 09-29	Jack added a second color sensor	Gym testing showed squaring on a line is impossible with one sensor
5	2026-09-29 (recommended)	Mini ball caster at the front	Almost no friction, turns best, small, sits below the body
6	2026-10-01	Two attachments adapted to the HummerOne bar	Lifting arm gained a seed bucket, 2-axis gained a hook
7	2026-10-01	Axle pins tried, quick pins kept	The lifting arm kept coming off with axle pins
8	2026-10-06	Gyro-based PID driving and turning blocks (Jordan)	Heading correction keeps the robot straight and turns accurate
9	2026-10-13	Technic ball joint replaced the mine cart wheels, Kp and Ki retuned	Better for turns and stays under the robot; the switch added wobble, fixed by retuning
10	2026-10-13	`Gyro Move` takes cm instead of degrees	Easier to communicate and test missions
11	2026-10-13	Seed-catching basket for Mission 11	Catches the seed, but it sometimes goes over the top
12	2026-10-06 to 10-15	Forklift adapted to the mounting bar	Lifts a platform; tallest attachment, needed a tipping test
13	2026-10-01	Lifting arm with seed bucket	Carries seeds in the order the seed mission needs
14	To confirm	Mission 1 attachment adapted	One of three options; the team picks one for final refinement
15	2026-10-01	2-axis attachment with hook	One input gives vertical and horizontal motion, and the hook adds a second job

The six scoring criteria (entry 2)

Criterion	HummerOne	Comp Bot	GummyBears / Droid Bot M	StarterBot
Archetype match to proven competitive designs	4.5	2	4	2.5
Attachment swap speed at base	4	3	3	2
Squaring (flat back and sides)	4	5	3	2
Stability under load	4	4.5	3	2
Maneuverability in tight areas	3	2	4	4
Drive repeatability (for PID code)	3.5	4	3	2
Total (of 30)	**23**	**20.5**	**20**	**14.5**

HummerOne's worst score was 3. Comp Bot beat it on squaring and repeatability but was much lower on archetype match and attachment swap. Our scores differed from the pre-meeting review (which had Comp Bot first at 26). Where we disagreed, the team's own view carried. We didn't buy StarterBot ($30): it bought nothing the free options didn't cover, and its sides are not flat.

Key facts by decision

Front support (entries 3, 5, 9). Three versions. (1) Original wheels have no rubber, so they skid or drag differently from run to run. (2) Mine cart wheels: lower profile, less friction when turning, very good in a straight line, but resisted when turning and stuck out past the sides, so a wheel could touch a wall before the body. (3) **Technic ball joint**: better for turns, stays under the robot, sides are clear for squaring. It added **wobble**, which Jordan fixed by retuning Kp and Ki. Guidance followed: smaller wheels wobble less and accuracy matters more than speed. Known risk: a caster can drift on hard acceleration and when reversing (**"it is being fixed"**).

Second color sensor (entry 4). Squaring on a line needs two sensors: when one sees black, the robot turns until the other also sees black, then wiggles until straight. Without squaring it drove crooked, and with squaring it didn't (our worksheet answer). Why it matters: setup mistakes at the table stop costing points once the robot straightens itself on a line. The square-up program was finished 2026-10-01.

Gyro blocks (entry 8).

- **`Gyro Move`** drives straight by correcting heading error every 0.1 seconds: proportional (Kp) plus integral (Ki), with an acceleration ramp (start at 15, speed up by 5, slow down by 8 at 70% of the distance).
- **`Gyro Turn`** turns to an angle at a speed that falls as the robot nears the target (25 to 90 power), and stops 3 degrees early so it doesn't overshoot.
- Both are reusable My Blocks, so any mission program can call them with a distance, a speed or an angle.

- Why yaw: yaw turns the robot by a real angle, while 90 motor degrees only turns the wheel a quarter turn.

Distance in cm (entry 10). The robot drove a measured 100 cm (1 m measuring stick) and that took 2070 motor degrees, so **20.7 degrees per cm**. About 5.75 wheel turns. Measuring on the table is better than calculating from the wheel size, because it includes tire squash and slip.

Attachment pins (entry 7). Jack suggested axle pins because it would be **faster to put on and off**. The lifting arm kept coming off, so we went back to **quick pins, which hold more securely**. A secure fit matters more than swap speed, because an attachment that falls off mid-run loses points.

Attachment swap speed. Our review put HummerOne's full-width bar at about 5 seconds to swap versus 20 seconds for a pinned design.

Attachments, what changed from the original builds (entries 12 to 15):

Attachment	Original build	What we changed	Why
Lifting arm (seed bucket)	`Lifting-Arm-Attachment.pdf`, a geared hinge arm for scoop and hook missions	Added a seed bucket. Quick pins instead of axle pins.	"We wanted to make it so we could bring the seed back and the plain geared hinge arm didn't work well." Seeds must go in a specific order.
2-axis movement (hook)	`2-Axis-Movement-Attachment.pdf`, bevel gears give vertical and horizontal motion from one input	Added a hook. Swapped to opposite sides. Longer pole.	Hook: "more convenient for pulling down the door" (Mission 11). Swapped sides: "so that it works better". Longer pole: "so it would reach the seed".
Forklift	`Forklift1-zoptyk.pdf`, rack-and-pinion vertical lift for platform missions	Adapted to HummerOne's mounting bar. Stabilized after it tipped.	Raises the platform (Robot Man's mission). Tallest attachment, biggest tipping risk.
Mission 1 attachment	To confirm	To confirm	Three options built from Giraffes #3388, Platinum Pizzeria and Team TurtleBots. The team picks one.

Attachment ratings from our worksheets (1 to 5):

Attachment	Pros	Cons	Rating
Lifting arm + seed bucket	"Masher good" **[?]**	"Not high enough"	3.5
2-axis + hook	"It works"	"gears fall off"	**5**
Forklift	"Its cool"	"Swinging it gets stuck" **[?]**	3.5

Mission 11 seed basket (entry 11). Removing the seed is worth **30 points**, but we want to **collect** it for another mission.

Idea tried	Catches out of 10	What happened
Walls on the side	3	"it fell out a lot"
Change the code	5	"didn't go high enough"
Bowl / basket	**7**	"it also fell out a lot"

D. Worksheet results (2026-10-29)

Full transcription: robot_design/worksheet-results-2026-10-29.md. Measurements were written in **mm** (the sheets asked for cm). **[?]** means hard to read.

D1. Kp and Ki tuning (bare robot)

Runner and measurer: Jackathon. Recorder: Fedor. Coder: Jordan. Battery 100%.

Step 1: baseline at the old values (Kp 2.1, Ki 2), 5 runs. Misses: 12.6, 7.4, 14, 5.5, 8.6 mm. The five numbers average about **9.6 mm** (just under 1 cm). All 5 passed.

Step 2a: Kp sweep with Ki = 0 (offset in mm, + means right, wobble score 0 to 3)

Kp	Run 1	Run 2	Run 3	Average miss
1.0	+23.9	+26.1	+29.4	26
1.5	+19	+23	+2.8	15
2.0	+1.8	0	0	**0.6**
2.5	+10.6	+5.4	15.7	about 10.6
3.0	+10.7	+5.9	23.3	13 (one wobble of 3)

Chosen Kp: 2.

Step 2b: Ki sweep with Kp = 2

Ki	Run 1	Run 2	Run 3	Average miss
0	+6.6	n/a	n/a	
0.5	+12.8	+6.6	+20.4	
1	0	+1.6	0	**under 1 mm**
2	0	+17.5	−2	6 mm

Chosen Ki: 1. Reason written: "Error was lowest".

What it shows

- **Kp 2 with Ki 1 is the best pair we tested.** Three runs ended within 1.6 mm of straight, versus about 9.6 mm for the old Kp 2.1 and Ki 2.

- **Low Kp drifts, high Kp weaves.** Kp 1.0 and 1.5 ended 19 to 29 mm off to the right. Kp 3.0 had a wobble score of 3 on one run.
- **The data is noisy.** Some values had runs of 17 to 20 mm next to runs near 0. Three runs per value is not enough to rank them closely. The 10-run confirmation at Kp 2 and Ki 1 is still to do.
- **Nearly every miss was to the right.** That could mean a steady pull right on the bare robot. Check for a wheel or ball joint problem.

D2. Attachments (baseline only, started)

Both sheets used Kp 2 and Ki 1 (the new values). Only 2 runs each.

Attachment	Run	Miss	Wobble
Forklift	1	3.5 mm	1
Forklift	2	not written	1
2-axis	1	3.6 mm	1
2-axis	2	8.8 mm	1

(The forklift sheet also has "−12 mm over 100 cm" and "−13.2 mm", which may be distance error rather than sideways error. Check.)

D3. Drive 1 meter (100 cm), ten times

Which Kp and Ki were used is not written.

Run	Off by
1	7.2 mm
2	2.9 mm
3	−4.0 mm
4	−3.6 mm
5	9.1 mm
6	"Perfect"
7 to 9	"less than 2"
10	"Perfect"

The numbers seem to get better over the 10 runs. Is that the robot, or the person measuring? Be ready for a judge to ask. Left blank: ten 90° turns and the ruler squaring check.

D4. Our written answers (in our own words)

Question	What we wrote
Why HummerOne?	"Hummer had the most good qualities like, its stability."
What do you remember about building it?	"I remembers modifying the robot's mini wheels because we didn't have the right ones."
Which scoring criterion mattered most?	"Drive repeatability matters the most to me. It is important to me and Ella because we need consistency."
Why was cost left out?	"Because we didn't care about the cost." (Better to say: we wanted to choose on how the robot performs in a match, and decide on cost separately.)
Why mine cart wheels?	"Cuz they fit, and the better option was too tall." Sylvie changed them, in September.
Before and after squaring	"Without the squaring it drove crooked and with the squaring it didn't."
Why "mini" ball caster?	"Cuz it could do more stuff and could get closer to certain missions."
Did it drift?	"Yes, it is being fixed."
Did the forklift tip?	"Yes, and we stabilized it so it's good now!"
What did Jack hope axle pins would do?	"It would be faster to put on off."
Describe the wobble	"When the robot is unstable the robot starts wiggling and wobbling."
Which mission does the hook do?	"The hook does Mission 11 and hooks on to the door."

D5. Photos and sketches

Four photos were taken on 2026-10-29: the **passive Mission 1 attachment** (top view), the **2-axis arm**, the **forklift** (tall black mast with forks and a gear on top), and the **seed bucket** (magenta tray on a yellow arm with a long black rod). Notebook sketches of the original 2-axis and lifting arm builds are in `robot design additional pages.pdf`. Bring these to judging to show the team's work (judges can see them even without the field).